

Lab: Tokenization, Lemmatization

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Topics to cover

- Comments: Assignment 1
- Project proposal
- Tokenization, Lemmatization

Comments: Assignment 1

- Overall great work!
- Put all files into a single `.zip` file.
- Exercise 4: **Suppose you are given a text file called** `sentences.txt`.
- Exercise 5: Please reflect my feedback into project proposal.

Project proposal

Project proposal

- Deadline: 9/18 11:59PM
- Guidelines:
`https://hksung.github.io/Fall26_SLS775/assignments/2_project%20proposal`
- One page is ideal.

Project proposal

1. Research Question(s)

Provide 1–2 focused research question(s), which should be:

- connected to a linguistic phenomenon or construction
- answerable using the data available to you
- appropriate for an NLP-based analysis

2. Corpus

Describe the corpus you plan to use.

- For many of you, this section can build on Assignment 1.
- If you plan to collect or compile your own corpus, focus on describing the domain representativeness of your corpus.

3. Unit of Analysis and Linguistic Measures

Describe what will count as one observation in your analysis. Include:

- your primary unit of analysis
- the linguistic feature(s), variable(s), or measure(s) you plan to examine at that level
- conceptually, how those features will be identified or measured in your NLP pipeline

This part is related to what we covered today. For the NLP pipeline section, I especially appreciate the (hand-drawn) illustration of the overall logic.

Project proposal

4. Mini Annotated Bibliography

Include at least 3 relevant sources. For each source:

- provide the full citation
- explain how the study is relevant to your project

(Optional 5). Comments/Questions

If you have comments/questions related to project, include here.

Evaluation

- 30 points (3% of the final grade)
- Please refer to the detailed rubric in Lamakū.

Tokenization, Lemmatization

Tasks

- **Set up the NLP tools**
 - Install spaCy
 - the English pipeline and Hugging Face Transformers
- **Explore tokenization**
 - Tokenize English texts with spaCy
 - Inspect token boundaries and token attributes
- **Explore lemmatization and morphological analysis**
 - Examine tokens, lemmas, POS tags, and morphological information
- **Compare linguistic and subword tokenization**
 - Tokenize English with BERT
 - Compare spaCy tokens with model-specific subword tokens.
- **Calculate corpus frequencies**
 - Extract tokens and lemmas from a text file
 - Calculate token/lemma frequencies, types, and hapax types
 - Save frequency lists as CSV files

More descriptions

- <https://hksung.github.io/PythonTutorial/5.html>
- Exercise 6 is the part of Assignment 2.